

non-differentiated and can be used as an open source. Instead of falling into it and just consuming it, we can use the best it offers,” says Dr Aloknath De, CTO, Samsung India.

The concept of collaborative development on networks and new technologies is not new in telecom. “Samsung uses the Tizen platform to build digital appliances and Jerryscript from javascript intended to run on a very constrained kind of environment. It also supports IoT and open connectivity foundations,” Aloknath says. Moreover, when it comes to 5G communication, he considers open random access network (O-RAN) and open networking automation platform (ONAP) as two big elements. Akraino is also an important element that supports high availability of cloud services, spanning a variety of use cases bringing artificial intelligence.

Fundamental shift in architecture

Industries are looking at product consumption, architecture software, modularity of software network functions, application services, and use of open source, showcasing a major shift in architecture.

Subodh Gajare, Lead Architect (5G and IoT Security), Cisco R&D, feels that nowhere between 2G and 4G did we witness such major architectural shifts. He says, “When we look at 5G on a silo compared to three previous generations of mobility, we see a fundamental shift in architecture in the way network components were looked at and the security element that has been blown apart.”

In specificity to open source Gajare adds, “So it (open source) is something that starts all from the client and merges on the telco DC side. We are witnessing three major architectural shifts, and this is the place where 7 trillion devices to the 7 trillion people and all these economies of scale will be met. And that’s why it’s both a huge competi-

tive advantage and a huge secondary challenge as well.”

Innovation in open network ecosystem

Gnanapriya Chidambaranathan, AVP & Senior Principal Architect, Infosys, says open networking is bringing in innovation, embracing the cloud nativeness, and dynamic orchestration and automation of network slices, bringing closed loop assurance and exposing open APIs for ecosystem integration.

For instance, all of us are connected remotely and enterprise workloads are moving from the data centres to the edge. Similarly, when we talk about the industry verticals, whether it is manufacturing or Industry 4.0, a lot of low-latency analytics and insights are required. And from a consumer perspective, users can watch high-definition video live streaming and the immersive experiences.

“There are a variety of possibilities that exist today. Open source brings in that openness and helps in driving innovation, cost-efficiently,” she says. RANs enable physical access to devices and were mostly developed as complete proprietary solutions. This means that it was difficult for innovations to happen at the same pace as the rest of the market.

“We are looking at the The Linux Foundation solution set, ORAN ecosystems, open stack and Kubernetes solutions. There is also the Open Networking Foundation (ONF) where there are a set of open source platforms that exist. We are also looking at how DLT can be leveraged for smart contracts as the business models and monetisation is also important,” Gnanapriya states.

Creating revolution

Open source has been a revolution over time and solutions like Linux and Android are part of our life already. According to experts, it is playing a key role in the next phase of revolutionising the data centre market. For instance, a lot of infrastructure for hyperscalers like Google

and Facebook runs on open source.

“We may not be aware that there are companies selling commercial networking products using open source to the data centre and cloud market. The next phase of the data centre evolution is the orchestration with OpenStack, Kubernetes,” says Abhijit Chaudhury, Founder and CEO, Niral Networks.

Acknowledging the pivotal role of 5G, IoT, edge and telecom, Abhijit says there are multiple initiatives for open source disaggregation happening in the 5G for RAN, Core, and Transport. “There have been a few smaller commercial 5G deployments using open source, but I think five/six years down the line there will be huge momentum towards open source based commercialisation in the telecom and the private cellular network for industrial/IoT use cases,” he adds.

Disaggregating components

When 5G came into existence, the thought process of disaggregating the components had already begun. With the arrival of open RAN and the disaggregated components talking to each other on open interfaces, the possibility of doing a lot of telecom applications in RAN and transporting domain to become smaller components that can be handled in isolation opened up.

“These are components that you can innovate upon and bring a lot of value into before you put them together to create a larger solution. So, from the service provider perspective, these are great developments,” says Manish Gangey, SVP and Head, R&D, Airtel. He also points out that one of the big problems in India is that we have a low ARPU (average revenue per user) in comparison to any other place. “So, the way we need to build our own infrastructure has to be thought about differently for any company to be profitable.”

Open source fits beautifully in an association like this by reducing the overall cost of ownership, accelerating innovation, and bringing in many